BACSE202 – DATABASE SYSTEMS

Project Details

on

AgriSmart: Smart Farm & Crop Recommendation System

Prepared by

Aarav Gupta - 25BAI0131

Chirag Saxena - 25BCE0223

Suyash Agarwal - 25BBS0088

Project Description

AgriSmart recommends the most suitable crop for a farmer's plot using low-cost soil sensors instead of a laboratory N-P-K test. An Arduino Uno node with a capacitive soil-moisture sensor, a DHT11 air temperature/humidity sensor and a DS18B20 soil-temperature probe sends readings over USB to a Python serial bridge, which validates them and stores them in a MySQL 8 database. Soil pH is optional and can be typed in from the Soil Health Card. The database has 9 tables in 3NF (farmers, plots, crops, sensor_nodes, soil_health_logs, cultivation_cycles, treatments, recommendation_runs, recommendation_items), 8 views, 3 triggers (crop-rotation rule, DS18B20 fault cleaning, node last-seen), 4 stored procedures (sp_start_cycle, sp_record_harvest, sp_plot_history, sp_yield_report) and the function fn_virtual_n_index. A Flask website lets a farmer register a field, see live readings, get plain-language crop advice and record what was sown, fertilised and harvested. The crop model is trained on the public Kaggle Crop Recommendation dataset (2,200 rows, 22 crops); N, P, K and rainfall were dropped, and soil moisture, soil temperature and soil type were derived with documented formulas. The deployed, isotonic-calibrated models are a Soft Voting ensemble (Random Forest + Hist Gradient Boosting + SVM) when pH is known and a Random Forest when it is not (Table 1).

Feature set	Deployed model	5-fold CV accuracy	Hold-out accuracy	Hold-out top-3
Sensors + pH	Soft Voting (RF+HGB+SVM)	0.895	0.902	0.993
Sensors only (no pH)	Random Forest	0.831	0.814	0.986

Table 1. Deployed models (seed 42; hold-out = 440 test rows, after isotonic calibration).

Novelty of the Idea

The primary novelty of this work lies in **moving the crop decision from a single sensor snapshot to the plot's stored history in the database**, so that a cheap node without an N-P-K sensor can still give reliable, explainable advice. Existing crop recommenders classify one reading, need N-P-K values, always return an answer even when unsure, and ignore what was grown on the plot before. AgriSmart addresses each of these with five contributions:

- **N1 – Quality-Weighted Temporal Consensus (QW-TC):** instead of the latest reading, the engine averages the last 7 days of readings, weighting each by a quality score (range check, DS18B20 fault codes −127/85 °C, a median-based spike test and soil-air temperature consistency) and a day-blocked recency weight. Under 0–30 % injected sensor faults it keeps top-1 accuracy at 0.857–0.902, against 0.539–0.614 for the latest reading alone (Figure 1).
- **N2 – Adaptive model switch when pH is missing:** if no valid pH (at most 180 days old) exists, a separately trained sensor-only model is used instead of guessing the pH. This gives 0.814 accuracy, compared with 0.791 when pH is filled with the training median and 0.677 when pH is assumed to be 7.0.

- **N3 – Calibrated, abstention-aware advice:** isotonic calibration lowered the expected calibration error from 0.086 to 0.022 (with pH) and from 0.074 to 0.051 (without pH). An Isolation-Forest detector flags 70.1 % of random in-range inputs but only 0.45 % of real test readings, and a 0.50 confidence threshold keeps 97.0 % of recommendations at 91.8 % accuracy, asking for a lab test for the rest.
- **N4 – Virtual Nitrogen Balance Index (VNBI) and context re-ranking:** with no N-P-K sensor, soil nitrogen is estimated from the plot's last three crops and the fertiliser recorded in the database (decay 0.6 per cycle), implemented both as the SQL function fn_virtual_n_index and in Python. The final score multiplies the model probability by season, rotation and nitrogen factors; a trigger blocks an annual crop from following itself.
- **N5 – Explainable fit card:** for every crop, the P5–P95 range of each measurement is learned from the training data and stored in the crops table. The website shows whether each measured value is inside, below or above that range, and the view v_recommended_crops uses the same ranges as a rule-based baseline that can be checked by hand.

Contribution	Experiment	Result
N1 QW-TC	Exp 3: 42 simulated readings over 7 days per test row, 0–30 % faults	Top-1: 0.902 / 0.875 / 0.857 at 0 / 10 / 30 % faults (latest reading only: 0.580 / 0.589 / 0.539)
N2 Model switch	Exp 4: test rows with pH removed	Sensor-only model 0.814; pH = median 0.791; pH = 7.0 0.677; pH measured 0.902
N3 Calibration + abstention	Exp 2 calibration; Exp 5 risk-coverage and OOD test	ECE 0.086 → 0.022; OOD flags 70.1 % random vs 0.45 % real; threshold 0.50 keeps 97.0 % at 91.8 %
N4 VNBI re-ranking	Exp 6: one soil reading, five plot histories	Last crop maize → maize blocked, lentil #1 (0.30); VNBI −1.96 → maize 0.62 → 0.52; November → lentil 0.38 above maize 0.37
N5 Fit card	Implemented for all 22 crops	Ranges stored in the crops table and shown per recommendation

Table 2. Evidence for each contribution (experiments/EXPERIMENTS_SUMMARY.md).

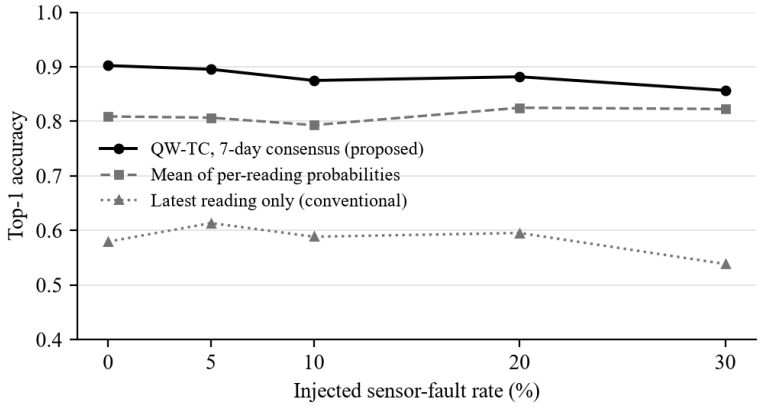


Figure 1. Top-1 accuracy versus injected sensor-fault rate (Exp 3, with-pH model, 440 test rows).

Limitations. Moisture, soil temperature and soil type in the training data are derived from a public dataset with documented formulas, not measured in the field, and the Exp 3 sensor windows are simulated. The VNBI weights and re-ranking factors are rule-based and are shown by case study, not learned from yield data. In the current prototype the DS18B20 probe is not yet fitted, so soil temperature is estimated from air temperature and moisture with the same formula used to build the training data.

Proposed Outcome: Conference